Resilient Neuroadaptive Distributed Fixed-Time Attitude Coordination Control for Multiple Spacecraft

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***Abstract*—This work studies the attitude coordination tracking problem for multiple spacecraft with consideration of unintended faults (communication link faults and actuator faults), inertial uncertainties, and external disturbances under a directed communication graph. A resilient neuroadaptive distributed fixed-time control scheme is investigated to solve this challenging problem. First, an improved adaptive distributed observer is established for followers to estimate the states of the leader when considering communication link faults. The proposed observer improves the resilience against communication link faults. Subsequently, to further cope with the problem of actuator faults, inertial uncertainties, and external disturbances, based on the proposed observer and the technique of adding a power integrator, a neuroadaptive distributed fixed-time attitude coordination controller is developed. Unlike the existing controllers, the proposed one requires less information when dealing with faults and lumped uncertainties, and has a lower-computational cost. Moreover, the fixed-time stability of the closed-loop system is ensured under the designed resilient neuroadaptive distributed control scheme. Finally, comparative simulations are carried out to manifest the effectiveness of the investigated coordination control method.**

***Index Terms*—Fixed-time control, multiple spacecraft, neuroadaptive, resilient control.**

I. Introduction

COMPARED with the conventional single spacecraft system, the multiple spacecraft system has the advantages of higher flexibility, reliability, and lower cost [1], [2]. Thus, the spacecraft formation flying (SFF) has been applied in a comprehensive range of complex space missions, for example, earth observation, astronomical observation, and deep space exploration. Due to the harsh and complex space environment, the resilient attitude coordination control, which implies that the controlled system can maintain a normal operational performance at a prescribed level in the existence of uncertainties, has attracted increasing attention in SFF. During the past decades, a significant amount of resilient attitude coordination control schemes have been carried out, for example, neural network (NN)-based control [3], [4], [5], robust control [6], [7], [8], and observer-based control [9], [10].

However, most of the aforementioned control methods are developed based on the assumption that no faults exist in the spacecraft system. As a matter of fact, such requirement may not always be met and unintended faults often occur in the practical spacecraft system. The existence of faults brings the significant degradation to the resilience of the multiple spacecraft control system [11]. For tackling the issue of faults, the fault tolerant control (FTC) has been developed. Overall, the available methods for FTC can be classified into two types: 1) the active one and 2) the passive one [12]. Owing to the less computation burden and the simpler structure, the passive FTC has been comprehensively used in the spacecraft attitude control system [13], [14], [15], [16], [17], [18]. For instance, in [13] and [15], the adaptive distributed attitude FTC problem was investigated for the networked spacecraft with unknown multiplicative actuator faults. In [16], a sliding mode FTC has been employed for multiple spacecraft in the presence of faults as well as uncertainties. An event-triggered FTC attitude coordination controller with uncertainties and input saturation was developed in [17] for the spacecraft formation. In the field of spacecraft attitude control, the vast majority of fault-tolerant control results typically only consider the actuator fault, whereas the problem of communication link faults among different spacecraft is not considered. In practice, communication link faults will result in a postponed and incorrect command transmission in SFF, which will lead to more severe consequences than actuator faults. Regrettably, since almost all the research on communication link faults focuses on the linear system, it is nontrivial to expand the existing works (e.g., [19], [20], [21], [22], and [23]) to the realization of the attitude coordination control problem. To date, there are few resilient attitude coordination control methods that can simultaneously address the problem of actuator and communication link faults.

Another practical factor in SFF that requires attention is the convergence rate. From the practical view, a fast convergence rate is essential to meet the requirement of a science objective in SFF. A typical weakness in all of the abovementioned solutions is that the fast convergence rate of the closed-loop system cannot be guaranteed; only the asymptotic or finite-time controllers are investigated. Fortunately, the fixed-time

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controller, which can provide the required convergence rate in a bounded time that being independent of the initial states, was proposed by Polyakov [24]. In this regard, the fixed-time approach has been utilized in various control systems [25], [26], [27], [28], [29], [30], [31]. In [25], a fixed-time attitude coordination control law was developed for SFF. Subsequently, to further enhance the resilience of the closed-loop system, a disturbance observer-based fixed-time attitude controller was investigated in [26]. Also, an output-feedback fixed-time attitude coordination problem was studied in [27]. It is noted, however, that most of the fixed-time solutions can only work under the undirected communication topology, which means that there exists a bidirectional information exchange between the adjacent spacecrafts. This is not the common case in reality. Meanwhile, it is still a challenging problem on how to develop a resilient fixed-time attitude coordination control scheme with consideration of unintended faults.

Motivated by the aforementioned observations, this work studies the resilient neuroadaptive distributed fixed-time attitude coordination control scheme under a directed topology. The problems of communication link faults, actuator faults, inertial uncertainties, and external disturbances are considered explicitly. A novel resilient observer is constructed for the formation member to obtain the leader spacecraft states, based on which a neuroadaptive fixed-time attitude coordination controller is devised. The major contributions of this work include three-fold.

1) A novel resilient distributed observer is first constructed for the formation member to access the leader spacecraft states in this work. Compared with the existing results [1], [4], [5], [9], [10], the communication link faults caused by the harsh and complex space environment are considered in our work. The adaptive gain technique is utilized to compensate for the effect of communication link faults in the observer design, which makes the developed observer more aerospace engineering orientated.
2) With the observed information, a neuroadaptive distributed fixed-time attitude controller is developed. In comparison with most resilient control schemes [13], [14], [15], [16], the proposed control scheme can be applied with less information, eliminating the need for the lower bound on multiplicative faults and the upper bound on lumped disturbances. Meanwhile, in comparison to the NN-based attitude control schemes [9], [14], [32], the adaptive parameters in the proposed controller are only scalars, which means lower-computational cost.
3) The rigorous fixed-time stability of the attitude control system is given by the Lyapunov approach under the devised resilient adaptive distributed control scheme. In comparison with [9] and [10], the convergence time is independent of the initial states, and the attitude tracking errors can converge to the origin with bounded errors. Moreover, compared to previous studies [26], [28], the observer and controller work simultaneously in this article. Therefore, the controller does not need to wait for the convergence of observation errors, which makes the proposed scheme much simpler for aerospace implementation.

The remaining part of this work is as follows. In Section II, the problem formulation is given, and some preliminaries are provided. Then, the resilient adaptive fixed-time attitude coordination control scheme is proposed in Section III, where the resilient observer and the fixed-time controller are developed. Illustrative examples are presented in Section IV. Section V draws some conclusive statements and future plans.

Notations: In this work, the following stand notations will be used. $\mathbb{R}$ denotes the real number, $\mathbb{R}^n$ denotes the n-dimensional Euclidean space, and $\mathbb{R}^{m\times n}$ denotes the set of $m \times n$ dimensional real matrices. For a vector $x = [x_1, x_2, \ldots, x_n]^T$ and a positive constant α, define $\mathrm{sig}^\alpha(x) = [\mathrm{sign}(x_1)|x_1|^\alpha, \mathrm{sign}(x_2)|x_2|^\alpha, \ldots, \mathrm{sign}(x_n)|x_n|^\alpha]^T$, where $\mathrm{sign}(\cdot)$ is the signum function, $\mathrm{diag}\{x_1, x_2, \ldots, x_n\}$ denotes the $n \times n$ dimensional diagonal matrix. The symbol 1_n denotes the $n \times 1$ column vector with all elements being ones, I_n denotes the n-dimensional unit matrix, $\|\cdot\|$ denotes the Euclidean norm or its induced norm, $\sigma_{\max}(\cdot)$ and $\sigma_{\min}(\cdot)$ stand for the maximum and minimum singular values of a matrix, respectively, $\otimes$ and $\circ$ denote the Kronecker product and the Hadamard product, respectively.

II. Problem Formulation and Preliminaries

A. Graph Theory

In this work, the leader-following multiple spacecraft system is considered, which consists of n follower spacecrafts and one leader spacecraft. The communication topology graph $G_s = (V, E, A)$ is utilized to describe the direct communication of the spacecraft formation, where $V = \{v_1, v_2, \ldots, v_n\}$ represents the node set, $E \subseteq V \times V$ represents the edge set, and $A = [a_{ij}] \in E$ represents the adjacency matrix. In G_s, the data flow is defined by a weight a_{ij} and an edge $\{v_j, v_i\}$ satisfying $a_{ij} = 1$ if $\{v_j, v_i\} \in E$, otherwise $a_{ij} = 0$. Let $N_i = \{j|(v_j, v_i) \in E\}$ be a set of neighbors of the follower spacecraft i, and $H = \mathrm{diag}\{h_i\} \in \mathbb{R}^{N\times N}$ be an in-degree matrix with $h_i = \sum_{j\in N_i} a_{ij}$. Then, the Laplacian matrix can be defined as $L = H - A$. If the leader spacecraft is a neighbor of the follower spacecraft i, then an edge (v_0, v_i) exists with a weight $b_i = 1$. Accordingly, the pinning matrix is defined as $B = \mathrm{diag}\{b_i\} \in \mathbb{R}^{N\times N}$. From follower spacecraft i to follower spacecraft j, a directed path is defined by a series of edges satisfying $[(v_i, v_k), (v_k, v_l), \ldots, (v_m, v_j)]$. When a node in the graph follows a directed path to every other node, the graph is said to have a spanning tree.

Assumption 1: The directed communication graph G_s of the spacecraft formation has a spanning tree, in which the leader spacecraft being its root.

B. Problem Formulation

In this work, the attitude kinematics and dynamics of the follower spacecraft are defined as

$$\dot{\sigma}_i = G_i(\sigma_i)\omega_i \tag{1}$$
$$J_i\dot{\omega}_i = -\omega_i^\times J_i\omega_i + T_i + d_i \tag{2}$$

where $G_i(\sigma_i) = (1/2)(\sigma_i^\times + \sigma_i\sigma_i^T + [(1-\sigma_i^T\sigma_i)/2]I_3)$, $\sigma_i = e_i\tan(\phi_i/4) \in \mathbb{R}^3$ is the modified Rodrigues parameters (MRPs) [33] describing the attitude regarding the inertial frame with $\phi_i \in [0, 2\pi)$rad and $e_i \in \mathbb{R}^3$ being the Euler eigenangle and eigenaxis, respectively. The vector $\omega_i \in \mathbb{R}^3$ denotes the angular velocity of the ith follower spacecraft in the body-fixed frame, the matrix $\omega_i^\times$ represents the skew-symmetric matrix of ω_i. The matrix $J_i = J_{0i} + J_{\Delta i} \in \mathbb{R}^{3\times3}$ denotes the inertia matrix of the ith follower spacecraft, in which $J_{0i} \in \mathbb{R}^{3\times3}$ is the nominal part that calibrated at the ground experiment stage and $J_{\Delta i} \in \mathbb{R}^{3\times3}$ is the unknown part. The vector $T_i \in \mathbb{R}^3$ denotes the control torque, which is produced by the reaction wheels (RWs) in this work. The vector $d_i \in \mathbb{R}^3$ is the unknown environmental disturbance torque.

Assumption 2: The external disturbance torque d_i caused by the complex spacecraft environment is assumed to be bounded. There exists an unknown positive constant $d_{\max}$ such that $\|d_i\| \le d_{\max}$.

Assumption 3: The desired attitude's first and second derivatives satisfy $\|\dot{\sigma}_0\| \le b_{\dot\sigma}$ and $\|\ddot{\sigma}_0\| \le b_{\ddot\sigma}$, where $b_{\dot\sigma} > 0$ and $b_{\ddot\sigma} > 0$ are known constants.

The unknown faults on the communication link can be expressed as [34]

$$\begin{cases} a_{fij}(t) = a_{ij} + f_{a_{ij}}(t) \\ b_{fi}(t) = b_i + f_{b_i}(t) \end{cases} \tag{3}$$

where a_{ij} and b_i denote the nominal weights in the communication link without faults, $f_{a_{ij}}(t)$ and $f_{b_i}(t)$ denote the time-varying faults, $a_{fij}(t)$, and $b_{fi}(t)$ denote the actual weights of the formation. Due to the communication fault model (3), the adjacency matrix becomes $A_f(t)$, the pinning matrix is given as $B_f(t)$. Correspondingly, the in-degree becomes the time-varying $D_f(t)$, and the Laplacian matrix becomes the time-varying $L_f(t)$.

Assumption 4: The communication faults $f_{aij}(t)$ and $f_{bi}(t)$ together with their first derivatives are bounded. There exist positive constants $a_{\max}$ and $b_{\max}$ such that $|f_{aij}(t)| \le a_{\max} < a_{ij}$ and $|f_{bi}(t)| \le b_{\max} < b_i$, which also means the signs of $a_{fij}(t)$ and $b_{fij}(t)$ are the same as the nominal ones.

Remark 1: In the existing spacecraft formation attitude control schemes, Assumption 3 is commonly used. The output of the spacecraft actuator is limited by actual physical factors and cannot produce infinite acceleration. The desired attitude is generated by an actual leader spacecraft. Thus, the assumption that the desired attitude's first and second derivatives are bounded is consistent with the realistic scenario. Moreover, the upper bounds can also be inferred as certain values in the ground test stage. In [35], angular accelerations are assumed to be exchangeable among the follower spacecrafts. In [36], the desired angular velocity is assumed to be available to all follower spacecrafts. Compared with the assumptions utilized in [35] and [36], Assumption 3 is less conservative. Assumption 4 is standard for the communication fault in the existing literature [20], [34], and [37]. Both the fading channel and the channel manipulation attack can be represented by the communication link fault model (3) under Assumption 4. Moreover, the connectivity of the nominal static communication graph is maintained with Assumption 4. Based on Assumption 4, one has the following property.

Property 1 [20]: There exist a positive-definite diagonal matrix $Q(t) = \mathrm{diag}\{q_1(t), q_2(t), \ldots, q_n(t)\}$ and a positive-definite matrix $P(t)$, such that $H^T(t)Q(t) + Q(t)H(t) = P(t)$, in which $H(t) = L_f(t) + B_f(t)$. In addition, $Q(t)$ together with its first derivative are bounded.

RWs' failures occur frequently due to the extreme space environment. The RWs mathematical model with multiplicative fault and additive fault can be established as follows [38], [39]:

$$T_i = \Gamma_i u_i + \bar{u}_i \tag{4}$$

where the time-varying diagonal matrix $\Gamma_i = \mathrm{diag}\{\Gamma_{i,1}, \Gamma_{i,2}, \Gamma_{i,3}\}$ represents the RWs effectiveness matrix with every element satisfying $(0, 1]$, $\bar{u}_i \in \mathbb{R}^3$ and $u_i \in \mathbb{R}^3$ denote the bounded additive actuator fault $\|\bar{u}_i\| \le \bar{u}_{\max}$ with $\bar{u}_{\max}$ being an unknown positive constant and the resilient controller, respectively.

By introducing $x_{1,i} = \sigma_i$ and $x_{2,i} = \dot{\sigma}_i$, the spacecraft attitude system equipped with RWs under actuator fault is represented as

$$\begin{cases} \dot{x}_{1,i} = x_{2,i} \\ \dot{x}_{2,i} = F_i(x_{1,i}, x_{2,i}) + G_i(x_{1,i})J_{0i}^{-1}(\Gamma_i u_i + \bar{u}_i + d_i) \end{cases} \tag{5}$$

where $F_i(x_{1,i}, x_{2,i}) = \dot{G}_i(x_{1,i})G_i^{-1}(x_{1,i})x_{2,i} - G_i(x_{1,i})J_{0i}^{-1}(G_i^{-1}(x_{1,i})x_{2,i})^\times J_i G_i^{-1}(x_{1,i}) \quad x_{2,i} - G_i(x_{1,i})J_{0i}^{-1}J_{\Delta i}(\dot{G}_i^{-1}(x_{1,i})x_{2,i} + G_i^{-1}(x_{1,i})\dot{x}_{i,2})$, $G_i(x_{1,i})$ is defined in (1).

Remark 2: MRPs attitude description has shadow set with a switch surface to achieve the global description of attitude motion subject to the constraint $\|\sigma\| \le 1$. A shadow counterpart of MPR vector $\sigma^S = -(\sigma/(\sigma^T\sigma))$ is used when $\sigma^T\sigma > 1$. By switching the MRPs to σ^S when $\sigma^T\sigma > 1$, the MRP's vector remains bounded within a unit sphere, global rotation representation without singularity can thus be ensured.

The objective of this work is to develop a resilient fixed-time distributed attitude coordination control scheme to ensure that the follower spacecraft can track the dynamic leader spacecraft under the direct graph. In general, the detailed control objectives are two-fold.

1) *Resilience to Communication Link Faults:* The follower spacecraft can obtain the states of the leader in the coexistence of unknown communication link faults.
2) *Resilience to Actuator Faults and Uncertainties:* The follower spacecraft can track the leader in a fixed time in the presence of actuator faults, inertial uncertainties, and external disturbances.

C. Preliminary Definitions

Consider the system

$$\dot{x}(t) = f(x(t)), x(0) = x_0 \tag{6}$$

where $x \in \mathbb{R}^n, f : \mathbb{R}^n \to \mathbb{R}^n$ is a continuous function. Suppose that the equilibrium point is the origin.

Lemma 1 [24] [40]: For the system described in (6). Suppose that there exist a positive definite Lyapunov

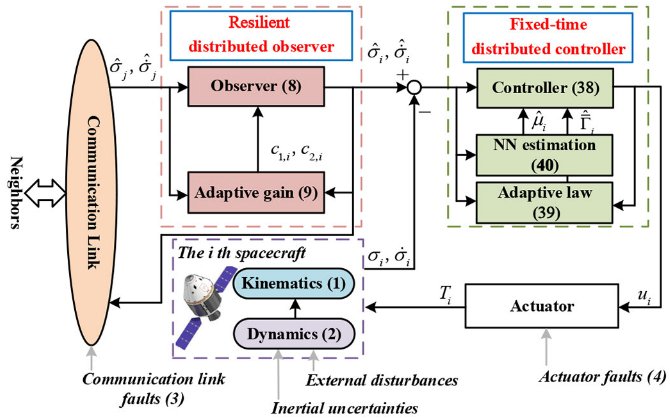

Fig. 1. Block diagram of the developed method.

function $V(x)$ and parameters $\psi_\alpha, \psi_\beta, p, q, k > 0$ with $kp < 1$, $kq > 1$, and $0 < \delta < \infty$, such that $\dot{V}(x) \leq -(\psi_\alpha V(x)^p + \psi_\beta V(x)^q)^k + \delta$. Then, the practical fixed-time stable of the system dynamics is ensured, namely, the state x will converge to the region $\{x | V(x) \leq \min\{\psi_\alpha^{-(1/p)}(\delta/1-\theta^k)^{(1/kp)}, \psi_\beta^{-(1/p)}(\delta/1-\theta^k)^{(1/kq)}\}\}$ with $0 < \theta < 1$ in a fixed time T, which is bounded by $T \leq [1/(\psi_\alpha^k \theta^k(1-kp))] + [1/(\psi_\beta^k \theta^k(kq-1))]$.

For a continuous nonlinear function $f(x) : \mathbb{R}^m \to \mathbb{R}^n$ in a compact set $\Omega_x \in \mathbb{R}^m$. There exists a radial basis function NN, such that

$$f(x) = W^{*T}\varphi(x) + \varepsilon \tag{7}$$

where $W^* \in \mathbb{R}^{l\times n}$ denotes the optimal weighting matrix with $l > 1$ standing for the number of the hidden node, $\varepsilon \in \mathbb{R}^n$ is the estimation error and satisfies $\|\varepsilon\| \leq \bar{\varepsilon}$ with $\bar{\varepsilon}$ being a small positive constant, $\varphi(x) = [\varphi_1(x), \ldots, \varphi_l(x)]^T$ is the basic function vector and chosen as the Gaussian function $\varphi_i(x) = \exp([((-x-\nu_i)^T(x-\nu_i))/(\rho_i^2)]), i = 1, \ldots, l$ with ν_i being the center of the receptive field and ρ_i being the width of the Gaussian function [41].

Lemma 2 [42] [43]: For any $x_i \in \mathbb{R}, i = 1, \ldots, n$, and a positive number ν, when $0 < \nu \leq 1$, then $(\sum_{i=1}^n |x_i|)^\nu \leq \sum_{i=1}^n |x_i|^\nu \leq n^{1-\nu}(\sum_{i=1}^n |x_i|)^\nu$; when $\nu > 1$, then $\sum_{i=1}^n |x_i|^\nu \leq (\sum_{i=1}^n |x_i|)^\nu \leq n^{\nu-1}\sum_{i=1}^n |x_i|^\nu$.

Lemma 3 [44]: For any $x, y \in \mathbb{R}$, and $0 < r \leq 1$, it holds that: $|\text{sig}^r(x) - \text{sig}^r(y)| \leq 2^{1-r}|x-y|^r$.

Lemma 4 [45]: For any $x, y \in \mathbb{R}$, $r_1, r_2 > 0$, and $p > 0$, it holds that: $|x|^{r_1}|y|^{r_2} \leq [(r_1 p)/(r_1+r_2)]|x|^{r_1+r_2} + [(r_2)/(p^{(r_1/r_2)}(r_1+r_2))]|y|^{r_1+r_2}$.

III. Resilient Adaptive Fixed-Time Control Scheme Design

A. Overview of the Developed Control Architecture

The block diagram of the developed control system is depicted in Fig. 1. The developed resilient adaptive fixed-time attitude coordination control system has an observer-controller architecture. In order to fulfill the control tasks, we first design the observer using the adaptive gain technique and then the controller by means of the neuroadaptive method.

B. Resilient Observer Design

Usually, for the follower spacecraft, a distributed state observer is employed since only a limited percentage of formation members can access the leader's information [25], [27]. However, the existing methods cannot be utilized to the communication link fault issue. To this end, a novel distributed observer with resilience to communication link faults is developed in this part. Denote the leader's attitude and angular velocity as $x_{1,0} = \sigma_0$ and $x_{2,0} = \dot{\sigma}_0$, and let $\hat{x}_{1,i}$ and $\hat{x}_{2,i}$ as the estimates of $x_{1,0}$ and $x_{2,0}$. Then, the distributed observer for the follower spacecrafts is given by

$$\begin{cases} \dot{\hat{x}}_{1,i} = -c_{1,i}^2\zeta_i - k\zeta_i - l_1\text{sign}(\zeta_i) \\ \dot{\hat{x}}_{2,i} = -c_{2,i}^2\psi_i - k\psi_i - l_2\text{sign}(\psi_i) \end{cases} \tag{8}$$

and the adaptive law

$$\begin{cases} \dot{c}_{1,i} = \zeta_i^T\zeta_i - r_1 c_{1,i} \\ \dot{c}_{2,i} = \psi_i^T\psi_i - r_1 c_{2,i} \end{cases} \tag{9}$$

where $i = 1, 2, \ldots, n$, $\zeta_i = \sum_{j=1}^n a_{fij}(t)(\hat{x}_{1,i}-\hat{x}_{1,j}) + b_{fi}(t)(\hat{x}_{1,i}-x_{1,0})$, $\psi_i = \sum_{j=1}^n a_{fij}(t)(\hat{x}_{2,i}-\hat{x}_{2,j}) + b_{fi}(t)(\hat{x}_{2,i}-x_{2,0})$. $k > 0$, $0 < \alpha < 1$, $\beta > 1$, $l_1 > b_{\dot{\sigma}}$, $l_2 > b_{\ddot{\sigma}}$, and $r_1 > 0$ are the observer gains. It should be noted that although the unknown faults $a_{fij}(t)$ and $b_{fi}(t)$ are in the estimation errors ζ_i and ψ_i, we do not need to resolve them directly. Actually, we can calculate the fixed-time observer output with the known corrupted information $a_{fij}(t)(\hat{x}_{1,i}-\hat{x}_{1,j})$, $b_{fi}(t)(\hat{x}_{1,i}-x_{1,0})$, $a_{fij}(t)(\hat{x}_{2,i}-\hat{x}_{2,j})$, and $b_{fi}(t)(\hat{x}_{2,i}-x_{2,0})$, which is standard for the distributed control with communication link faults. More detailed explanation can be founded in [20], [34], and [37]. Then, the first main theorem is presented as follows.

Theorem 1: Consider the attitude system (5) under the uncertain communication link with the designed distributed observer (8) and the adaptive law (9). If Assumptions 1, 3, and 4 hold, then $\hat{x}_{1,i}$ as well as $\hat{x}_{2,i}$ will converge to $x_{1,0}$ and $x_{2,0}$ with bounded errors.

Proof: Let $\bar{x}_1 = \hat{x}_1 - 1_n \otimes x_{1,0}$ and $\bar{x}_2 = \hat{x}_2 - 1_n \otimes x_{2,0}$ with $\hat{x}_1 = (\hat{x}_{1,1}^T, \ldots, \hat{x}_{1,n}^T)^T$ and $\hat{x}_2 = (\hat{x}_{2,1}^T, \ldots, \hat{x}_{2,n}^T)^T$. Define $\zeta = (\zeta_1^T, \ldots, \zeta_n^T)^T = (H(t) \otimes I_3)\bar{x}_1$ and $\psi = (\psi_1^T, \ldots, \psi_n^T)^T = (H(t)\otimes I_3)\bar{x}_2$. Bearing in mind that $H(t)$ is an invertible matrix, $\bar{x}_1 = 0$ and $\bar{x}_2 = 0$ are equivalent to $\zeta = 0$ and $\psi = 0$, respectively. Thus, if $\|\zeta\|$ and $\|\psi\|$ converge to the region near zero under the observer (8), then the leader spacecraft states $x_{1,0}$ and $x_{2,0}$ will be reconstructed with bounded errors.

According to the definition of ψ and (8), one has

$$\begin{aligned} \dot{\psi} &= (H(t) \otimes I_3)\dot{\bar{x}}_2 + \big(\dot{H}(t) \otimes I_3\big)\bar{x}_2 \\ &= -\Big(H(t)C_2^2 \otimes I_3\Big)\psi - k(H(t) \otimes I_3)\psi \\ &\quad -(H(t) \otimes I_3)(l_2\text{sign}(\psi) + 1_n \otimes \dot{x}_2) \\ &\quad +\big(\dot{H}(t) \otimes I_3\big)\bar{x}_2 \end{aligned} \tag{10}$$

where $C_2 = \text{diag}\{c_{2,1}, \ldots, c_{2,n}\}$.

Consider the following Lyapunov function candidate:

$$V_\psi = \sum_{i=1}^n q_i(t)\psi_i^T\psi_i + \sum_{i=1}^n \frac{1}{r_2}\big(c_{2,i} - \alpha_\psi\big)^2 \tag{11}$$

where $r_2 > 0$, α_ψ is a positive constant to be designed shortly, and $q_i(t)$ is positive number that defined in Property 1. Evidently, V_ψ is positive definite.

The time derivative of V_ψ is given as

$$\dot{V}_\psi = \sum_{i=1}^{n} \dot{q}_i(t)\psi_i^T\psi_i + 2\sum_{i=1}^{n} q_i(t)\psi_i^T\dot{\psi}_i + \sum_{i=1}^{n} 2r_2^{-1}(c_{2,i} - \alpha_\psi)\dot{c}_{2,i}. \quad (12)$$

Substituting (9) and (10) into (12) yields

$$\begin{aligned}\dot{V}_\psi \leq{} & \psi^T[\dot{Q}(t)\otimes I_3]\psi - 2\psi^T\left[Q(t)H(t)C_2^2\otimes I_3\right]\psi \\ & -k\psi^T[P(t)\otimes I_3]\psi \\ & -2\psi^T[Q(t)H(t)\otimes I_3](l_2\text{sign}(\psi) + 1_n\otimes\dot{x}_{2,0}) \\ & +2\psi^T[Q(t)\dot{H}(t)\otimes I_3]\bar{x}_2 \\ & +\sum_{i=1}^{n} 2r_2^{-1}(c_{2,i}-\alpha_\psi)(\psi_i^T\psi_i - r_1c_{2,i}) \end{aligned} \quad (13)$$

where Property 1 is utilized, $\dot{Q}(t) = \text{diag}\ \{\dot{q}_1(t), \dot{q}_2(t), \ldots, \dot{q}_n(t)\}$, and $P(t)$ is detailed in Property 1.

Note that $H(t)$ is a nonsingular and invertable matrix according to its definition. Hence, one has

$$\|\bar{x}_2\| \leq \frac{\|\psi\|}{\sigma_{\min}(H(t)\otimes I_3)}. \quad (14)$$

For the third term on the right-hand-side (RHS) of (13), there exists

$$\begin{aligned}& -2\psi^T[Q(t)H(t)\otimes I_3](L_2\text{sign}(\psi) + 1_n\otimes\dot{x}_{2,0}) \\ & = 2\sum_{i=1}^{n} q_i(t)\psi_i^T\Big(\sum\nolimits_{j=1}^{n} a_{fij}(t)(-l_2\text{sign}(\psi_i) + l_2\text{sign}(\psi_j)) \\ & +b_{fi}(t)(-l_2\text{sign}(\psi_i) - \dot{x}_{2,0})\Big) \leq 0. \end{aligned} \quad (15)$$

Under Assumption 3, if $\psi_{i,k} > 0, k = 1,2,3,$ there exists $(\sum_{j=1}^{n} a_{fij}(t)(-l_2\text{sign}(\psi_{i,k}) + l_2\text{sign}(\psi_{j,k})) + b_{fi}(t)\times\ (-l_2\text{sign}(\psi_{i,k}) - \dot{x}_{2,0,k})) \leq 0$. If $\psi_{i,k} < 0$, there exists $(\sum_{j=1}^{n} a_{fij}(t)(-l_2\text{sign}(\psi_{i,k}) + l_2\text{sign}(\psi_{j,k})) + b_{fi}(t)\times\ (-l_2\text{sign}(\psi_{i,k}) - \dot{x}_{2,0,k})) \geq 0$. If $\psi_{i,k} = 0$, there exists $\psi_{j,k}(\sum_{j=1}^{n} a_{fij}(t)(-l_2\text{sign}(\psi_{i,k}) + l_2\text{sign}(\psi_{j,k})) + b_{fi}(t)\times(-l_2\text{sign}(\psi_{i,k}) - \dot{x}_{2,0,k})) = 0$. Therefore, (15) is true.

The last term on the RHS of (13) can be expressed as

$$\begin{aligned}& 2r_2^{-1}(c_{2,i}-\alpha_\psi)(\psi_i^T\psi_i - r_1c_{2,i}) \\ & = 2r_2^{-1}c_{2,i}(\psi_i^T\psi_i) - 2r_2^{-1}\alpha_\psi(\psi_i^T\psi_i) \\ & -2r_1r_2^{-1}c_{2,i}^2 + 2r_1r_2^{-1}\alpha_\psi c_{2,i} \\ & \leq 2r_2^{-1}c_{2,i}(\psi_i^T\psi_i) - 2r_2^{-1}\alpha_\psi(\psi_i^T\psi_i) \\ & -r_1r_2^{-1}(c_{2,i}-\alpha_\psi)^2 + r_1r_2^{-1}\alpha_\psi^2. \end{aligned} \quad (16)$$

Then, substituting (14)–(16) into (13) leads to

$$\begin{aligned}\dot{V}_\psi \leq{} & -\lambda_0k\psi^T\psi \\ & -\left(2\frac{\alpha_\psi}{r_2} - \frac{\lambda_2}{\sigma_{\min}(H(t)\otimes I_3)} - 2\frac{c_{2,\max}}{r_2} - \lambda_1 - \lambda_3\right) \\ & \times\psi^T\psi - \sum_{i=1}^{n} r_1r_2^{-1}(c_{2,i}-\alpha_\psi)^2 + nr_1r_2^{-1}\alpha_\psi^2 \end{aligned} \quad (17)$$

where $c_{2,\max} = \max\{|c_{2,i}|\}$, $\lambda_0 = \sigma_{\min}(P(t))$, $\lambda_1 = \max_{\forall t\geq 0}\|\dot{Q}(t)\|$, $\lambda_2 = 2\max_{\forall t\geq 0}\|Q(t)\dot{H}(t)\|$, $\lambda_3 = 2\max_{\forall t\geq 0}\|Q(t)H(t)C_2^2\|$.

Define a compact set as $\Omega_{\psi,i} = \{\psi_i \| \psi_i| \leq \Xi_{\psi,i}\}$ with $\Xi_{\psi,i}$ being a positive constant. Based on the definition of $c_{2,i}$ in (9), if ψ_i on the compact $\Omega_{\psi,1}\times\Omega_{\psi,2}\times\cdots\times\Omega_{\psi,n}$, it can be obtained that there always exists a positive constant Λ_i such that $|c_{2,i}| \leq \Lambda_i$. Thus, it can always find an appropriate α_ψ such that $\alpha_\psi \geq [([(\lambda_2)/\ (\sigma_{\min}\ (H(t)\otimes I_3))]\ + 2r_2^{-1}c_{2,\max} + \lambda_1 + \lambda_3)/\ (2r_2^{-1})]$.

Then, (17) can be rewritten as

$$\begin{aligned}\dot{V}_\psi & \leq -\lambda_0k\psi^T\psi - \sum_{i=1}^{n} r_1r_2^{-1}(c_{2,i}-\alpha_\psi)^2 + nr_1r_2^{-1}\alpha_\psi^2 \\ & \leq -\eta_1\left[\sum_{i=1}^{n}\left(q_i(t)\psi_i^T\psi_i + r_2^{-1}(c_{2,i}-\alpha_\psi)^2\right)\right] \\ & \quad + nr_1r_2^{-1}\alpha_\psi^2 \\ & = -\eta_1V_\psi + \upsilon_1 \end{aligned} \quad (18)$$

where $\eta_1 = \min([\lambda_0k/q_{\max}], r_1)$ with $q_{\max} = \max\{q_i(t)\}$, and $\upsilon_1 = nr_1r_2^{-1}\alpha_\psi^2$.

In light of the boundedness theorem [46] and (18), $(c_{2,i} - \alpha_\psi)$ and ψ_i are uniformly ultimately bounded. Therefore, it is easy to obtain that there exists a positive constant $\Delta_{\psi 1}$ such that $\sum_{i=1}^{n}(c_{2,i}-\alpha_\psi)^2 \leq \Delta_{\psi 1}$ and $\sum_{i=1}^{n} q_i(t)\psi_i^T\psi_i \leq \Delta_{\psi 1}$.

Following the similar design in (11), another Lyapunov function candidate is considered:

$$V_\zeta = \sum_{i=1}^{n} q_i(t)\zeta_i^T\zeta_i + \sum_{i=1}^{n}\frac{1}{r_2}(c_{1,i}-\alpha_\zeta)^2 \quad (19)$$

where α_ζ is a positive constant to be designed. Based on the above development of V_ψ, we can obtain the following result: if α_ζ satisfies the related inequality, then $\hat{x}_{1,i} \to x_{1,0}$ with a bounded error $\Delta_{\zeta 1}$. Based on the above analysis, we can draw a conclusion that the proposed observer (8) can reconstruct the leader's states with bounded errors.

This completes the proof. ■

Based on the result of Theorem 1, the fixed-time stability of the designed observer can also be achieved.

Corollary 1: Consider the attitude system (5) under the uncertain communication link with the designed distributed observer (8) and the adaptive law (9). If Assumptions 1, 3, and 4 hold, then $\hat{x}_{1,i}$ as well as $\hat{x}_{2,i}$ will converge to $x_{1,0}$ and $x_{2,0}$ with bounded errors in a fixed time T_1, which is detailed later.

Proof: Note that

$$\begin{aligned}& -(c_{2,i}-\alpha_\psi)^2 \\ & = \Big[-\frac{1}{4}(c_{2,i}-\alpha_\psi)^2 - \frac{1}{4}(|c_{2,i}-\alpha_\psi| - |c_{2,i}-\alpha_\psi|^\alpha)^2 \\ & + \frac{1}{4}|c_{2,i}-\alpha_\psi|^{2\alpha} - \frac{1}{2}|c_{2,i}-\alpha_\psi|^{1+\alpha}\Big] \\ & + \Big[-\frac{1}{4}(c_{2,i}-\alpha_\psi)^2 - \frac{1}{4}(|c_{2,i}-\alpha_\psi| - |c_{2,i}-\alpha_\psi|^\beta)^2 \end{aligned}$$

$$+\frac{1}{4}|c_{2,i}-\alpha_\psi|^{2\beta}-\frac{1}{2}|c_{2,i}-\alpha_\psi|^{1+\beta}\Big]$$
$$\le -\frac{1}{4}\big(c_{2,i}-\alpha_\psi\big)^2+\frac{1}{4}|c_{2,i}-\alpha_\psi|^{2\alpha}-\frac{1}{4}\big(c_{2,i}-\alpha_\psi\big)^2$$
$$+\frac{1}{4}\big|c_{2,i}-\alpha_\psi\big|^{2\beta}-\frac{1}{2}\big|c_{2,i}-\alpha_\psi\big|^{1+\alpha}-\frac{1}{2}\big|c_{2,i}-\alpha_\psi\big|^{1+\beta}. \tag{20}$$

Then, by virtue of $(c_{2,i}-\alpha_\psi)^2 \le \Delta_{\psi 1}$ in Theorem 1, we can obtain that: when $(c_{2,i}-\alpha_\psi)^2 \ge 1$, $-(1/4)(c_{2,i}-\alpha_\psi)^2+(1/4)|c_{2,i}-\alpha_\psi|^{2\alpha}-(1/4)(c_{2,i}-\alpha_\psi)^2+(1/4)|c_{2,i}-\alpha_\psi|^{2\beta} \le (1/4)\Delta_{\psi 1}^{\beta} + (1/4)\Delta_{\psi 1}$; when $(c_{2,i}-\alpha_\psi)^2 < 1$, $-(1/4)(c_{2,i}-\alpha_\psi)^2+(1/4)|c_{2,i}-\alpha_\psi|^{2\alpha}-(1/4)(c_{2,i}-\alpha_\psi)^2+(1/4)|c_{2,i}-\alpha_\psi|^{2\beta} < (1/2)$. In summary, one has

$$-\big(c_{2,i}-\alpha_\psi\big)^2 \le -\frac{1}{2}\big|c_{2,i}-\alpha_\psi\big|^{1+\alpha}-\frac{1}{2}\big|c_{2,i}-\alpha_\psi\big|^{1+\beta}+\frac{1}{4}\Delta_{\psi 1}^{\beta}+\frac{1}{4}\Delta_{\psi 1}+\frac{1}{2}. \tag{21}$$

Similar to (21), one has

$$-\psi_i^T\psi_i \le -\frac{1}{2}\big(\psi_i^T\psi_i\big)^{\frac{\alpha}{2}+\frac{1}{2}}-\frac{1}{2}\big(\psi_i^T\psi_i\big)^{\frac{\beta}{2}+\frac{1}{2}}+\frac{1}{4}\frac{\Delta_{\psi 1}}{q_{\min}}+\frac{1}{4}\left(\frac{\Delta_{\psi 1}}{q_{\min}}\right)^{\beta}+\frac{1}{2} \tag{22}$$

where $q_{\min} = \min\{q_i(t)\}$.

By selecting the same Lyapunov function candidate in (11) and the appropriate α_ψ, utilizing (21), (22), and Lemma 2, the time derivative of V_ψ is given as

$$\begin{aligned}\dot V_\psi \le& -\frac{1}{2}\lambda_0 k\sum_{i=1}^{n}\left(\big(\psi_i^T\psi_i\big)^{\frac{\alpha+1}{2}}+\big(\psi_i^T\psi_i\big)^{\frac{\beta+1}{2}}\right)\\ &+n\lambda_0 k\left(\frac{1}{4}\frac{\Delta_{\psi 1}}{q_{\min}}+\frac{1}{4}\left(\frac{\Delta_{\psi 1}}{q_{\min}}\right)^{\beta}+\frac{1}{2}\right)\\ &-\frac{1}{2}r_1r_2^{-1}\sum_{i=1}^{n}\left(\big|c_2-\alpha_\psi\big|^{1+\alpha}+\big|c_2-\alpha_\psi\big|^{1+\beta}\right)\\ &+nr_1r_2^{-1}\alpha_\psi^2+nr_1r_2^{-1}\left(\frac{1}{4}\Delta_{\psi 1}^{\beta}+\frac{1}{4}\Delta_{\psi 1}+\frac{1}{2}\right)\\ \le& -\eta_2V_\psi^{\frac{1+\alpha}{2}}-\eta_3V_\psi^{\frac{1+\beta}{2}}+\upsilon_2\end{aligned} \tag{23}$$

where $\eta_2 = \min\{(1/2)\lambda_0k[1(q_{\max}^{(1+\alpha)/2})], (1/2)r_1r_2^{(1+\alpha/2)-1}\}$, $\eta_3 = 2^{1-(1+\beta/2)}\min\ \{(1/2)\lambda_0kn^{1-(1+\beta/2)}\ [1/(q_{\max}^{(1+\beta)/2})], (1/2)r_1r_2^{(1+\beta/2)-1}\ n^{1-(1+\beta/2)}\}$, and $\upsilon_2 = n\lambda_0k((1/4)[(\Delta_{\psi 1})/(q_{\min})] +(1/4)([(\Delta_{\psi 1})/(q_{\min})])^{\beta} + (1/2)) +nr_1r_2^{-1}\alpha_\psi^2 +nr_1r_2^{-1}((1/4)\Delta_{\psi 1}^{\beta}+ (1/4)\Delta_{\psi 1} + (1/2))$ with $q_{\max} = \max\{q_i(t)\}$.

According to Lemma 1, it follows from (23) that V_ψ is driven to converge into the region $D_1 = \{V_\psi \le \min\{\eta_2^{-(2/\alpha+1)}((\vartheta_2/1-\varsigma_1))^{(2/\alpha+1)}, \eta_3^{-(2/\alpha+1)}\ ((\vartheta_2/1-\varsigma_1))^{(2/\beta+1)}\}\}$ in the fixed time $T_{1,\psi} \le [1/(\eta_2\varsigma_1(1-(1+\alpha/2)))] + [1/(\eta_3\varsigma_1([(1+\beta)/2]-1))]$. Therefore, (23) implies that $\hat{x}_{2,i} \to x_{2,0}$ with a bounded error $\Delta_{\psi 2} = \|\hat{x}_{2,i}-x_{2,0}\| \le [(\sqrt{(D_1/\sigma_{\min}(Q(t)))})/(\sigma_{\min}(H(t)\otimes I_3))]$ in the fixed time $T_{1,\psi}$.

Based on the above development of V_ψ, we can obtain the following result for $V_\zeta = \sum_{i=1}^{n} q_i(t)\zeta_i^T\zeta_i + (1/r_2)\sum_{i=1}^{n}(c_{1,i}-\alpha_\zeta)^2$ with α_ζ being a positive constant: if α_ζ satisfies the related inequality, then $\hat{x}_{1,i} \to x_{1,0}$ with a bounded error $\Delta_{\zeta 2}$ in the fixed time $T_{1,\zeta}$. Under the above analysis, we can draw a conclusion that the proposed observer (8) can reconstruct the leader's states with bounded errors in the fixed time $T_1 = \max\{T_{1,\psi}, T_{1,\zeta}\}$.

This completes the proof. ∎

Remark 3: The time-varying communication link fault presents a challenge to the accurate and timely command transmission in SFF. To tackle this issue, a novel distributed observer is proposed. The main improvements of the proposed observer are three-fold: 1) by carefully designing the adaptive law (9) and the corresponding Lyapunov function candidates V_ψ and V_ζ, a modification is introduced in the design of adaptive law to exclude the strict increasing phenomenon of the adaptive gain $c_{1,i}$ and $c_{2,i}$ compared with the existing work [20], which further enhances the resilience of the designed observer with measurement noises; 2) compared to previous work [20], the observer designed in this article only requires the upper bound information of the leader's commands, without the need for specific parameters of the leader's dynamical equation. Thus, the proposed observer has better applicability; and 3) the parameters α_ψ and α_ζ are required to satisfy the related inequalities. Calculating these parameters is challenging because they are associated with the unknown communication topology. However, since α_ψ and α_ζ are not observer parameters, we do not need to specifically compute them. In contrast, in existing literature [34], the observer parameter must be computed to ensure the relevant inequality holds for observer convergence. Therefore, the designed observer is more convenient for engineering applications.

Remark 4: Based on the designed adaptive resilient observer, the fixed-time convergence of the observation errors is proved in Corollary 1. Such conclusion has not been yielded in the existing results [20], [34], and [37]. It is worth noting that the power terms associated with α and β are not introduced in the designed observer. This is because, when considering both the directed graph and time-varying communication faults, it is nontrival to select an appropriate Lyapunov function to prove fixed-time convergence.

C. Fixed-Time Controller Design

To cope with actuator faults and uncertainties and further increase the resilience of the multiple spacecraft attitude system, a neuroadaptive fixed-time attitude coordination controller is investigated in this part. Define the attitude and angular velocity tracking errors as $\tilde{x}_{1,i} = x_{1,i} - \hat{x}_{1,i}$ and $\tilde{x}_{2,i} = x_{2,i} - \hat{x}_{2,i}$, respectively. From (5) and (8), the derivatives of these errors with respect to time are given as follows:

$$\begin{cases}\dot{\tilde{x}}_{1,i} = \dot{x}_{1,i} - \dot{\hat{x}}_{1,i}\\ \qquad = \tilde{x}_{2,i} + c_{1,i}^2\zeta_i + k\zeta_i + l_1\mathrm{sign}(\zeta_i)\\ \dot{\tilde{x}}_{2,i} = \dot{x}_{2,i} - \dot{\hat{x}}_{2,i}\\ \qquad = F_i\big(x_{1,i}, x_{2,i}\big) + G_i\big(x_{1,i}\big)J_{0i}^{-1}(\Gamma_iu_i + \bar{u}_i + d_i)\\ \qquad\quad + c_{2,i}^2\psi_i + k\psi_i + l_2\mathrm{sign}(\psi_i).\end{cases} \tag{24}$$

Next, we will utilize a recursive backstepping approach to design the controller.

Step 1: Consider the following Lyapunov candidate:

$$V_{1,i} = \frac{1}{2}\tilde{x}_{1,i}^T\tilde{x}_{1,i}. \tag{25}$$

Differentiating $V_{1,i}$ along with (24) yields

$$\dot{V}_{1,i} = \tilde{x}_{1,i}^T\tilde{x}_{2,i} + \tilde{x}_{1,i}^T\Big(c_{1,i}^2\zeta_i + k\zeta_i + l_1\mathrm{sign}(\zeta_i)\Big). \tag{26}$$

Define $\tilde{x}_{2,i}^* = \tilde{x}_{2,i} + p_2\mathrm{sig}^{\sigma}(\tilde{x}_{1,i})$ and denote $x_{2,i}^* = -p_1\mathrm{sig}^{\theta}(\tilde{x}_{1,i})$ as the virtual control law with $p_1 > 0$, $p_2 > 0$, $0.5 < \theta < 1$, and $\sigma > 1$ being the designed parameters. We show the first term on the RHS of (26) satisfies $\tilde{x}_{1,i}^T\tilde{x}_{2,i} = \sum_{k=1}^3 \tilde{x}_{1,ik}(\tilde{x}_{2,ik}^* - x_{2,ik}^*) - p_1\sum_{k=1}^3|\tilde{x}_{1,ik}|^{1+\theta}$ $-p_2\sum_{k=1}^3|\tilde{x}_{1,ik}|^{1+\sigma}$. By utilizing Lemmas 3 and 4, one has

$$\begin{aligned}\sum_{k=1}^3\tilde{x}_{1,ik}\big(\tilde{x}_{2,ik}^* - x_{2,ik}^*\big) &\le \sum_{k=1}^3 2^{1-\theta}|\tilde{x}_{1,ik}||\eta_{ik}|^{\theta}\\ &\le \sum_{k=1}^3 2^{1-\theta}\left(\frac{1}{1+\theta}|\tilde{x}_{1,ik}|^{1+\theta} + \frac{\theta}{1+\theta}|\eta_{ik}|^{1+\theta}\right)\end{aligned} \tag{27}$$

where $\eta_{ik} = \mathrm{sig}^{(1/\theta)}(\tilde{x}_{2,ik}^*) - \mathrm{sig}^{(1/\theta)}(x_{2,ik}^*)$. By utilizing Lemma 4, the second term on the RHS of (26) yields

$$\begin{aligned}\sum_{k=1}^3\tilde{x}_{1,ik}\chi_{ik} &\le \sum_{k=1}^3|\tilde{x}_{1,ik}||\chi_{ik}|\\ &\le \sum_{k=1}^3\left(\frac{1}{1+\sigma}|\tilde{x}_{1,ik}|^{1+\sigma} + \frac{\sigma}{1+\sigma}|\chi_{ik}|^{1+\frac{1}{\sigma}}\right)\end{aligned} \tag{28}$$

where $\chi_{ik} = c_{1,i}^2\zeta_{ik} + k\zeta_{ik} + l_1\mathrm{sign}(\zeta_{ik})$. Then, substituting (27) and (28) into (26) gives

$$\begin{aligned}\dot{V}_{1,i} \le &-\left(p_2 - \frac{1}{1+\sigma}\right)\sum_{k=1}^3|\tilde{x}_{1,ik}|^{1+\sigma} - \left(p_1 - \frac{2^{1-\theta}}{1+\theta}\right)\\ &\times\sum_{k=1}^3|\tilde{x}_{1,ik}|^{1+\theta} + \frac{\sigma}{1+\sigma}\sum_{k=1}^3|\chi_{ik}|^{1+\frac{1}{\sigma}}\\ &+\frac{2^{1-\theta}\theta}{1+\theta}\sum_{k=1}^3|\eta_{ik}|^{1+\theta}.\end{aligned} \tag{29}$$

Step 2: On the basis of the adding one power integrator technique, we choose the following Lyapunov function candidate as:

$$V_{2,i} = \frac{2^{\theta-1}}{2-\theta}\sum_{k=1}^3\int_{x_{2,ik}^*}^{\tilde{x}_{2,ik}^*}\mathrm{sig}^{2-\theta}\left(\mathrm{sig}^{\frac{1}{\theta}}(s) - \mathrm{sig}^{\frac{1}{\theta}}\left(x_{2,ik}^*\right)\right)ds. \tag{30}$$

The derivative of (30) with respect to time is given as

$$\begin{aligned}\dot{V}_{2,i} = &-2^{\theta-1}\sum_{k=1}^3\frac{d\mathrm{sig}^{\frac{1}{\theta}}\left(x_{2,ik}^*\right)}{dt}\\ &\times\int_{x_{2,ik}^*}^{\tilde{x}_{2,ik}^*}\mathrm{sig}^{1-\theta}\left(\mathrm{sig}^{\frac{1}{\theta}}(s) - \mathrm{sig}^{\frac{1}{\theta}}\left(x_{2,ik}^*\right)\right)ds\\ &+\frac{2^{\theta-1}}{(2-\theta)}\sum_{k=1}^3\mathrm{sig}^{2-\theta}(\eta_{ik})\dot{\tilde{x}}_{2,ik}^*.\end{aligned} \tag{31}$$

For the first term on the RHS of (31), one has

$$\begin{aligned}&-2^{\theta-1}\frac{d\mathrm{sig}^{\frac{1}{\theta}}\left(x_{2,ik}^*\right)}{dt}\times\\ &\in t_{x_{2,ik}^*}^{\tilde{x}_{2,ik}^*}\mathrm{sig}^{1-\theta}\left(\mathrm{sig}^{\frac{1}{\theta}}(s) - \mathrm{sig}^{\frac{1}{\theta}}\left(x_{2,ik}^*\right)\right)ds\\ &\le 2^{\theta-1}p_1^{\frac{1}{\theta}}\left|\tilde{x}_{2,ik} + \chi_{ik}\right|\left|\tilde{x}_{2,ik}^* - x_{2,ik}^*\right||\eta_{ik}|^{1-\theta}\\ &\le p_1^{\frac{1}{\theta}}\left(\left|\tilde{x}_{2,ik}\right| + |\chi_{ik}|\right)|\eta_{ik}|\\ &\le \left(p_1^{\frac{1}{\theta}}2^{1-\theta} + \frac{p_1^{2+\theta+\frac{1}{\theta}}}{1+\theta} + p_1^{\frac{1}{\theta}}\frac{1}{1+\theta}\right)\times|\eta_{ik}|^{1+\theta}\\ &+\frac{\theta}{1+\theta}\left|\tilde{x}_{1,ik}\right|^{1+\theta} + \frac{\sigma}{1+\sigma}\left|\tilde{x}_{1,ik}\right|^{1+\sigma}\\ &+\frac{\left(p_1^{\frac{1}{\theta}}p_2\right)^{1+\sigma}}{1+\sigma}|\eta_{ik}|^{1+\sigma} + p_1^{\frac{1}{\theta}}\frac{\theta}{1+\theta}|\chi_{ik}|^{1+\frac{1}{\theta}}\end{aligned} \tag{32}$$

where Lemmas 3 and 4 and $[(d\mathrm{sig}^{(1/\theta)}(x_{2,ik}^*))/dt] = [(d\mathrm{sig}(-p_1^{(1/\theta)}\mathrm{sig}^{(1/\theta)}(\tilde{x}_{1,ik})^{\theta}))/(d\tilde{x}_{1,ik})][(d\tilde{x}_{1,ik})/dt] = -p_1^{(1/\theta)}(\tilde{x}_{2,ik} + \chi_{ik})$ are used. For the second term on the RHS of (31), one has

$$\dot{\tilde{x}}_{2,ik}^* = \dot{\tilde{x}}_{2,ik} + \frac{d\big(p_2\mathrm{sig}^{\sigma}\left(\tilde{x}_{1,ik}\right)\big)}{dt}. \tag{33}$$

Substituting (32) and (33) into (31) yields

$$\begin{aligned}\dot{V}_{2,i} \le &\sum_{k=1}^3\left(\left(p_1^{\frac{1}{\theta}}2^{1-\theta} + \frac{p_1^{2+\theta+\frac{1}{\theta}}}{1+\theta} + p_1^{\frac{1}{\theta}}\frac{1}{1+\theta}\right)|\eta_{ik}|^{1+\theta}\right.\\ &+\frac{\theta}{1+\theta}\left|\tilde{x}_{1,ik}\right|^{1+\theta} + \frac{\sigma}{1+\sigma}\left|\tilde{x}_{1,ik}\right|^{1+\sigma}\\ &\left.+\frac{\left(p_1^{\frac{1}{\theta}}p_2\right)^{1+\sigma}}{1+\sigma}\times|\eta_{ik}|^{1+\sigma} + p_1^{\frac{1}{\theta}}\frac{\theta}{1+\theta}|\chi_{ik}|^{1+\frac{1}{\theta}}\right)\\ &+\frac{2^{\theta-1}}{(2-\theta)}\times\left(\mathrm{sig}^{2-\theta}(\eta_i)\right)^T\Big(F_i\big(x_{1,i},x_{2,i}\big)\\ &+G_i\big(x_{1,i}\big)J_{0i}^{-1}(\Gamma_iu_i + \bar{u}_i + d_i) + c_{2,i}^2\psi_i + k\psi_i\\ &+l_2\mathrm{sign}(\psi_i) + p_2\sigma\left|\tilde{x}_{1,i}\right|^{\sigma-1}\circ\dot{\tilde{x}}_{1,i}\Big)\end{aligned} \tag{34}$$

where (5) and (24) are used.

Step 3: The neuroadaptive fixed-time attitude controller is developed in this step. For the convenience of the controller design, define a new failure indicator matrix $\Upsilon_i = \mathrm{diag}\{\Upsilon_{i1}, \Upsilon_{i2}, \Upsilon_{i3}\}$ with $(I_3 - \Upsilon_i)u_i = \Gamma_iu_i$, and $\bar{\Upsilon}_i = [\Upsilon_{i1}, \Upsilon_{i2}, \Upsilon_{i3}]^T$ of the failure indicator Υ_i that satisfying

$$\Upsilon_iu_i = \mathrm{diag}(u_i)\bar{\Upsilon}_i. \tag{35}$$

Note that the uncertainty $F_i(x_{1,i}, x_{2,i})$ in (34) can be approximated by the NN system (7), namely

$$F_i\big(x_{1,i}, x_{2,i}\big) = W_i^{*T}\varphi_i\big(x_{1,i}, x_{2,i}\big) + \varepsilon_i \tag{36}$$

where W_i^*, $\varphi_i(x_{1,i}, x_{2,i})$, and ε_i are the ideal constant weight, basic function vector, and NN estimation error of the ith spacecraft, respectively. According to Assumption 2, (4), (7), and $\|G_i(x_{1,i})\| \leq (1/2)$ in [3] and [33], one has $\|\varepsilon_i + G_i(x_{1,i})J_{0i}^{-1}(\bar{u}_i + d_i)\| \leq \bar{\varepsilon} + (1/2)\|J_{0i}^{-1}\|\|\bar{u}_{\max} + d_{\max}\|$. Therefore, one has

$$\left\| F_i\left(x_{1,i}, x_{2,i}\right) + G_i\left(x_{1,i}\right)J_{0i}^{-1}(\bar{u}_i + d_i)\right\| \leq \mu_i \|\Phi_i\| \tag{37}$$

where $\mu_i = \max_{t\geq 0} \|[W_i^{*T}, \varepsilon_i + G_i(x_{1,i})J_{0i}^{-1}(\bar{u}_i + d_i)]\|$ and $\Phi_i = [\varphi_i(x_{1,i}, x_{2,i})^T, 1]^T$.

According to (35)–(37), the attitude controller is developed as

$$\begin{aligned} u_i = &-\left(I_3 - \hat{\Upsilon}_i\right)^{-1}\left(G_i\left(x_{1,i}\right)J_{0i}^{-1}\right)^{-1}\Bigg(c_{2,i}^2\psi_i + k\psi_i \\ &+ l_2\mathrm{sign}(\psi_i) + p_2\sigma\left|\tilde{x}_{1,i}\right|^{\sigma-1} \circ \dot{\tilde{x}}_{1,i} \\ &+ k_1\mathrm{sig}^{2\theta-1}(\eta_i) + k_2\mathrm{sig}^{\sigma+\theta-1}(\eta_i) \\ &+ \frac{\hat{\mu}_i\|\Phi_i\|}{\left\|\mathrm{sig}^{2-\theta}(\eta_i)\right\|}\mathrm{sig}^{2-\theta}(\eta_i)\Bigg) \end{aligned} \tag{38}$$

$$\dot{\hat{\bar{\Upsilon}}}_i = -k_3\left(G_i\left(x_{1,i}\right)J_{0i}^{-1}(\mathrm{diag}(u_i))\right)^T\mathrm{sig}^{2-\theta}(\eta_i) - k_4\hat{\bar{\Upsilon}}_i \tag{39}$$

$$\dot{\hat{\mu}}_i = k_5\left\|\mathrm{sig}^{2-\theta}(\eta_i)\right\|\|\Phi_i\| - k_6\hat{\mu}_i \tag{40}$$

where $\hat{\Upsilon}_i$ is the estimate of the failure indicator matrix, $\hat{\bar{\Upsilon}}_i = [\hat{\Upsilon}_{i1}, \hat{\Upsilon}_{i2}, \hat{\Upsilon}_{i3}]^T$ is the estimate of the indicator vector $\bar{\Upsilon}_i$, $\hat{\mu}_i$ is the estimate of μ_i, $k_i(i = 1, 2, \ldots, 6)$ are the appropriate tuning parameters. Then, the second main theorem is presented as follows.

Theorem 2: Given the multiple spacecraft system, including a leader spacecraft and n follower spacecrafts described by (5). Suppose Assumptions 1–4 hold. If the attitude control law is developed as (38) with $\hat{\Upsilon}_i$ updated by (39) and $\hat{\mu}_i$ updated by (40), the control parameters satisfy $p_1 > [(2^{1-\theta})/(1+\theta)] + (\theta/1+\theta)$, $p_2 > 1$, $k_1 > [(2^{1-\theta}\ (\theta/1+\theta) + p_1^{(1/\theta)}2^{1-\theta} - [(p_1^{2+\theta+(1/\theta)})/(1+\theta)] - p_1^{(1/\theta)}(1/1+\theta))/(2^{1-\theta}(2-\theta))]$, $k_2 > [((2^{1-\theta}(2-\theta))^{-1}(p_1^{(1/\theta)}p_2)^{1+\sigma})/(1+\sigma)]$, $k_3 > 0$, $k_4 > 0$, $k_5 > 0$, and $k_6 > 0$, then the practically fixed-time tracking of the attitude system (5) can be guaranteed in the presence of communication link faults, actuator faults, inertial uncertainties, and external disturbances.

Proof: Choose the Lyapunov function candidate as

$$V_{c,i} = V_{1,i} + V_{2,i} + \frac{2^{\theta-1}}{2(2-\theta)k_5}\tilde{\mu}_i^2 + \frac{2^{\theta-1}}{2(2-\theta)k_3}\tilde{\bar{\Upsilon}}_i^T\tilde{\bar{\Upsilon}}_i \tag{41}$$

where $V_{1,i}$ and $V_{2,i}$ are defined in (25) and (30), respectively, $\tilde{\bar{\Upsilon}}_i = [\tilde{\Upsilon}_{i1}, \tilde{\Upsilon}_{i2}, \tilde{\Upsilon}_{i3}]^T = \bar{\Upsilon}_i - \hat{\bar{\Upsilon}}_i$ represents the estimation error of $\bar{\Upsilon}_i$, $\tilde{\mu}_i = \mu_i - \hat{\mu}_i$ is the estimation error of μ_i. By utilizing the former results ((29) and (34)) and the designed controller (38), the time derivative of $V_{c,i}$ is

$$\begin{aligned} \dot{V}_{c,i} \leq &-\bar{q}_1\sum_{k=1}^{3}\left|\tilde{x}_{1,ik}\right|^{1+\theta} - \bar{q}_2\sum_{k=1}^{3}\left|\tilde{x}_{1,ik}\right|^{1+\sigma} \\ &- \bar{q}_3\sum_{k=1}^{3}|\eta_{ik}|^{1+\theta} - \bar{q}_4\sum_{k=1}^{3}|\eta_{ik}|^{1+\sigma} + \bar{q}_5\sum_{k=1}^{3}|\chi_{ik}|^{1+\frac{1}{\theta}} \\ &+ \bar{q}_5\sum_{k=1}^{3}|\chi_{ik}|^{1+\frac{1}{\sigma}} + \frac{2^{\theta-1}}{2-\theta}\left(\mathrm{sig}^{2-\theta}(\eta_i)\right)^T \\ &\times\left(F_i\left(x_{1,i}, x_{2,i}\right) + G_i\left(x_{1,i}\right)J_{0i}^{-1}(\bar{u}_i + d_i)\right. \\ &\left.-G_i\left(x_{1,i}\right)J_{0i}^{-1}\mathrm{diag}(u_i)\tilde{\bar{\Upsilon}}_i\right) - \frac{2^{\theta-1}}{2-\theta}\hat{\mu}_i\|\Phi_i\| \\ &\times\left\|\mathrm{sig}^{2-\theta}(\eta_i)\right\| - \frac{2^{\theta-1}}{(2-\theta)k_5}\tilde{\mu}_i\dot{\hat{\mu}}_i - \frac{2^{\theta-1}}{(2-\theta)k_3}\tilde{\bar{\Upsilon}}_i^T\dot{\hat{\bar{\Upsilon}}}_i \end{aligned} \tag{42}$$

where $\bar{q}_1 = p_1 - [(2^{1-\theta})/\ (1+\theta)] - [\theta/(1+\theta)]$, $\bar{q}_2 = p_2 - [1/(1+\sigma)] - [\sigma/(1+\sigma)]$, $\bar{q}_3 = [k_1/\ (2^{1-\theta}(2-\theta))] - 2^{1-\theta}\ (\theta/1+\theta) - p_1^{(1/\theta)}\ 2^{1-\theta} + [(p_1^{2+\theta+\frac{1}{\theta}})/(1+\theta)] + p_1^{(1/\theta)}(1/1+\theta)$, $\bar{q}_4 = (k_2/2^{1-\theta}(2-\theta)) - [((p_1^{(1/\theta)}p_2)^{1+\sigma})/(1+\sigma)]$, $\bar{q}_5 = (\sigma/1+\sigma) + p_1^{(1/\theta)}(\theta/1+\theta)$.

Substituting (39) and (40) into (42) leads to

$$\begin{aligned} \dot{V}_{c,i} \leq &-\bar{q}_1\sum_{k=1}^{3}\left|\tilde{x}_{1,ik}\right|^{1+\theta} - \bar{q}_2\sum_{k=1}^{3}\left|\tilde{x}_{1,ik}\right|^{1+\sigma} - \bar{q}_3\sum_{k=1}^{3}|\eta_{ik}|^{1+\theta} \\ &- \bar{q}_4\sum_{k=1}^{3}|\eta_{ik}|^{1+\sigma} + \bar{q}_5\sum_{k=1}^{3}|\chi_{ik}|^{1+\frac{1}{\theta}} + \bar{q}_5\sum_{k=1}^{3}|\chi_{ik}|^{1+\frac{1}{\sigma}} \\ &- \bar{q}_6\tilde{\mu}_i^2 - \bar{q}_7\tilde{\bar{\Upsilon}}_i^T\tilde{\bar{\Upsilon}}_i + \bar{q}_7\bar{\Upsilon}_i^T\bar{\Upsilon}_i + \bar{q}_6\mu_i^2 \end{aligned} \tag{43}$$

where $\tilde{\bar{\Upsilon}}_i^T\hat{\bar{\Upsilon}}_i \leq -(1/2)\tilde{\bar{\Upsilon}}_i^T\tilde{\bar{\Upsilon}}_i + (1/2)\bar{\Upsilon}_i^T\bar{\Upsilon}_i$, $\tilde{\mu}_i\hat{\mu}_i \leq -(1/2)\tilde{\mu}_i^2 + (1/2)\mu_i^2$, and $(\mathrm{sig}^{2-\theta}(\eta_i))^T\ (F_i(x_{1,i}, x_{2,i}) + G_i(x_{1,i})J_{0i}^{-1}(\bar{u}_i + d_i)) \leq \mu_i\|\mathrm{sig}^{2-\theta}(\eta_i)\|\|\Phi_i\|$ are used, $\bar{q}_6 = [(2^{\theta-1}k_6)/(2(2-\theta)k_5)]$, and $\bar{q}_7 = [(2^{\theta-1}k_4)/(2(2-\theta)k_3)]$.

From the boundedness theorem [46] and (43), we can easily get that $\tilde{\mu}_i$ and $\tilde{\bar{\Upsilon}}_i$ are uniformly ultimately bounded, namely, $\tilde{\mu}_i^2 \leq \Delta_\mu$ and $\tilde{\bar{\Upsilon}}_i^T\tilde{\bar{\Upsilon}}_i \leq \Delta_\Upsilon$, where Δ_μ and Δ_Υ are some positive constants. Consequently, based on the same development as in (20), we have the following inequalities:

$$-\tilde{\mu}_i^2 \leq -\frac{1}{2}\left|\tilde{\mu}_i\right|^{1+\theta} - \frac{1}{2}\left|\tilde{\mu}_i\right|^{1+\delta} + \frac{1}{4}\Delta_\mu^\delta + \frac{1}{4}\Delta_\mu + \frac{1}{2} \tag{44}$$

$$\begin{aligned} -\tilde{\bar{\Upsilon}}_i^T\tilde{\bar{\Upsilon}}_i \leq &-\frac{1}{2}\left(\tilde{\bar{\Upsilon}}_i^T\tilde{\bar{\Upsilon}}_i\right)^{\frac{1+\theta}{2}} - \frac{1}{2}\left(\tilde{\bar{\Upsilon}}_i^T\tilde{\bar{\Upsilon}}_i\right)^{\frac{1+\delta}{2}} \\ &+ \frac{1}{4}\Delta_\Upsilon^\delta + \frac{1}{4}\Delta_\Upsilon + \frac{1}{2}. \end{aligned} \tag{45}$$

Next, substituting (44) and (45) into (43) leads to

$$\begin{aligned} \dot{V}_{c,i} \leq &-\bar{q}_1\sum_{k=1}^{3}\left|\tilde{x}_{1,ik}\right|^{1+\theta} - \bar{q}_2\sum_{k=1}^{3}\left|\tilde{x}_{1,ik}\right|^{1+\sigma} - \bar{q}_3\sum_{k=1}^{3}|\eta_{ik}|^{1+\theta} \\ &- \bar{q}_4\sum_{k=1}^{3}|\eta_{ik}|^{1+\sigma} - \frac{1}{2}\bar{q}_6\left|\tilde{\mu}_i\right|^{1+\theta} - \frac{1}{2}\bar{q}_6\left|\tilde{\mu}_i\right|^{1+\delta} \\ &- \frac{3^{\frac{\theta-1}{2}}}{2}\bar{q}_7\sum_{k=1}^{3}\left|\tilde{\bar{\Upsilon}}_{ik}\right|^{1+\theta} - \frac{1}{2}\bar{q}_7\sum_{k=1}^{3}\left|\tilde{\bar{\Upsilon}}_{ik}\right|^{1+\delta} + \bar{M}_{1,i} \end{aligned} \tag{46}$$

where Lemma 2 is used, $\bar{M}_{1,i} = (1/4)\bar{q}_6\Delta_\mu^\delta + (1/4)\bar{q}_6\Delta_\mu + (1/2)\bar{q}_6 + (1/4)\bar{q}_7\Delta_\Upsilon^\delta + (1/4)\bar{q}_7\Delta_\Upsilon + (1/2)\bar{q}_7 + \bar{q}_7\bar{\Upsilon}_i^T\bar{\Upsilon} + \bar{q}_6\mu_i^2 + \bar{q}_5\sum_{k=1}^{3}|\chi_{ik}|^{1+(1/\sigma)} + \bar{q}_5\sum_{k=1}^{3}|\chi_{ik}|^{1+(1/\theta)}$.

Based on the conditions of the control parameters, one has

$$\dot{V}_{c,i} \leq -\min\left\{\bar{q}_1, \bar{q}_3, \frac{1}{2}\bar{q}_6, \frac{3^{\frac{\theta-1}{2}}}{2}\bar{q}_7\right\} \times P_{1,i}$$

$$-\min\left\{\bar{q}_2, \bar{q}_4, \frac{1}{2}\bar{q}_6, \frac{1}{2}\bar{q}_7\right\} \times P_{2,i} + \bar{M}_{1,i} \quad (47)$$

where

$$P_{1,i} = \sum_{k=1}^{3}\left(\left|\tilde{x}_{1,ik}\right|^{1+\theta} + |\eta_{ik}|^{1+\theta} + \left|\tilde{\bar{\Upsilon}}_{ik}\right|^{1+\theta}\right) + \left|\tilde{\mu}_i\right|^{1+\theta}$$

$$P_{2,i} = \sum_{k=1}^{3}\left(\left|\tilde{x}_{1,ik}\right|^{1+\sigma} + |\eta_{ik}|^{1+\sigma} + \left|\tilde{\bar{\Upsilon}}_{ik}\right|^{1+\delta}\right) + \left|\tilde{\mu}_i\right|^{1+\delta}.$$

Noting that $V_{1,i} + V_{2,i} \le (1/2)\tilde{x}_{1,i}^T\tilde{x}_{1,i} + [(2^{\theta-1})/(2-\theta)]\sum_{k=1}^{3}|\tilde{x}_{2,ik}^* - x_{2,ik}^*||\eta_{ik}|^{2-\theta} \le (1/2)\tilde{x}_{1,i}^T\tilde{x}_{1,i} + [1/(2-\theta)]\sum_{k=1}^{3}|\eta_{ik}|^2 \le [1/(2-\theta)]\sum_{k=1}^{3}(|\tilde{x}_{ik}|^2 + |\eta_{ik}|^2)$, one has $V_{c,i} \le E(\sum_{k=1}^{3}(|\tilde{x}_{ik}|^2 + |\eta_{ik}|^2 + |\tilde{\bar{\Upsilon}}_{ik}|^2) + \tilde{\mu}_i^2)$ with $E = \max\{[1/(2-\theta)], [(2^{\theta-1})/(2(2-\theta)k_5)], [(2^{\theta-1})/(2(2-\theta)k_3)]\}$. Invoking Lemma 2 again, one has

$$V_{c,i}^{\frac{\theta+1}{2}} \le E^{\frac{\theta+1}{2}}P_{1,i}, V_{c,i}^{\frac{\sigma+1}{2}} \le E^{\frac{\sigma+1}{2}}10^{\frac{\sigma+1}{2}-1}P_{2,i}. \quad (48)$$

Substituting (48) into (47) yields

$$\dot{V}_{c,i} \le -Q_1 V_{c,i}^{\frac{\theta+1}{2}} - Q_2 V_{c,i}^{\frac{\sigma+1}{2}} + \bar{M}_{1,i} \quad (49)$$

where $Q_1 = [(\min\{\bar{q}_1, \bar{q}_3, (1/2)\bar{q}_6, [(3^{[(\theta-1)/2]})/2]\bar{q}_7\})/(E^{[(\theta+1)/2]})]$, $Q_2 = [(\min\{\bar{q}_2, \bar{q}_4, (1/2)\bar{q}_6, (1/2)\bar{q}_7\})/(E^{(\sigma+1/2)}10^{(\sigma+1/2)-1})]$.

Because the boundness of ζ_i and $c_{1,i}$ is guaranteed in Theorem 1. Therefore, it is straightforward that there always exists an upper bound for the last two terms $\bar{q}_5\sum_{k=1}^{3}|\chi_{ik}|^{1+(1/\sigma)} + \bar{q}_5\sum_{k=1}^{3}|\chi_{ik}|^{1+(1/\theta)}$ on the RHS of $\bar{M}_{1,i}$. Denote the least upper bound by $N_i = \sup(\bar{q}_5\sum_{k=1}^{3}|\chi_{ik}|^{1+(1/\sigma)} + \bar{q}_5\sum_{k=1}^{3}|\chi_{ik}|^{1+(1/\theta)})$. In light of Lemma 1 and (49), $\tilde{x}_{1,ik}$ and η_{ik} will converge to a region D_2:$\|\tilde{x}_{1,i}\|, \|\eta_i\|\,|V_{c,i} \le \min\{Q_1^{(-2/\theta+1)}([(\bar{M}_{1,i})/(1-\varsigma_2)])^{[2/(\theta+1)]}, Q_2^{[-2/(\theta+1)]}([(\bar{M}_{1,i})/(1-\varsigma_2)])^{(2/\sigma+1)}\}$ within a fixed time $T_{2,i}$ that estimated by $T_{2,i} \le [1/(Q_1\varsigma_2(1-[(\theta+1)/2]))] + [1/(Q_2\varsigma_2([(\sigma+1)/2]-1))]$ with $0 < \varsigma_2 < 1$. According to the definition of $V_{c,i}$, one can easily obtain that $\|\tilde{x}_{1,i}\|$ as well as $\|\tilde{x}_{2,i}\|$ converge to the region D_2 within the fixed time $T_{2,i}$. It holds that there exists a $T = \max\{T_{2,i}\}$ such that the practical fixed-time tracking can be achieved.

This completes the proof. ■

Remark 5: Different from the existing control schemes, the proposed controller can be implemented with less demanding information. On one hand, by resorting to the equivalent transformation $\Upsilon_i u_i = \mathrm{diag}(u_i)\bar{\Upsilon}_i$ and the adaptive law (39), the lower bound of the multiplicative fault is no longer required in the design or/and the analysis procedure compared to the previous works [13], [14], [15], [16]. On the other hand, in contrast to the work [47], the proposed controller does not require exact information of lumped uncertainties, that is, a known upper bound. The unknown upper bound is estimated by the adaptive NN technique (40). Such improvements are deemed favorable in the practical aerospace application.

Remark 6: In the conventional NN-based control schemes [4], [9], all of the elements in the weight matrix W are updated in the controller design, which leads to an extensive computation. By approximating the upper bound of $\|F_i(x_{1,i}, x_{2,i}) + G_i(x_{1,i})J_{0i}^{-1}(\bar{u}_i + d_i)\|$, only a scalar instead of a matrix related to the NN is updated, which reduces the computation burden. Moreover, in comparison with the available attitude FTC schemes, no fault detection and diagnosis are needed for the proposed method to solve the problem of actuator faults.

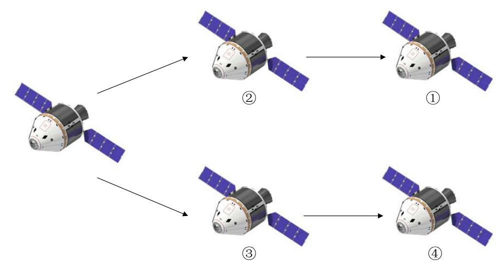

Fig. 2. Communication topology.

Remark 7: Both actuator faults and communication faults are considered in this article. Thus, the problem we consider is more complex and closer to the real mission than the problem studied in the existing result [18]. The proposed method can also be seen as an extension of the result studied in [20], which is a control scheme for synchronization control that only considers communication link faults. Moreover, compared with [18] and [20], the scheme we studied can provide a fixed-time stability result.

Remark 8: Compared to the existing NN-based FTC [14], the proposed scheme has two main innovations: 1) it addresses both communication and actuator faults simultaneously, which has not been done in the previous result and 2) it has lower-computational requirement compared to the existing method. Thus, the proposed approach has higher-practical engineering value.

IV. Simulation

Simulation results are carried out in this section to show the effectiveness of the developed resilient control scheme. Specifically, the formation considered in this section is made up of a leader spacecraft with four follower spacecrafts. Fig. 2 illustrates the topology of the communication graph.

The nominal parts of the inertia matrix for the follower spacecrafts are assumed to be $J_{01} = [11, 0, 0; 0, 10, 0; 0, 0, 11]$ kg · m^2, $J_{02} = [12, 0, 0; 0, 11, 0; 0, 0, 11]$ kg · m^2, $J_{03} = [12, 0, 0; 0, 10, 0; 0, 0, 12]$ kg · m^2, and $J_{04} = [10, 0, 0; 0, 9, 0; 0, 0, 11]$ kg · m^2, respectively. The uncertain parts of the inertia matrix for the follower spacecrafts are assumed to be $J_{\Delta 1} = [0, 04, 0.2; 0.4, -1, 0.6; 0.2, 0.6, 0]$ kg · m^2, $J_{\Delta 2} = [1, 0.3, 0.1; 0.2, 0, 0.5; 0.1, 0.5, 0]$ kg · m^2, $J_{\Delta 3} = [1, 0.3, 0.1; 0.2, -1, 0.5; 0.1, 0.5, 1]$ kg · m^2, and $J_{\Delta 4} = [-1, 0.3, 0.1; 0.2, -2, 0.5; 0.1, 0.5, 0]$ kg · m^2, respectively.

The initial values of the follower spacecrafts are set as $\sigma_1(0) = [0.3, 0.2, -0.35]^T$, $\sigma_2(0) = [0.3, 0.2, 0.35]^T$, $\sigma_3(0) = [0.3, -0.2, -0.35]^T$, $\sigma_4(0) = [-0.3, 0.2, -0.35]^T$ and $\dot{\sigma}_1(0) = [-0.1, 0.1, -0.1]^T$, $\dot{\sigma}_2(0) = [0.1, -0.1, 0.1]^T$, $\dot{\sigma}_3(0) = [0.1, 0.1, -0.1]^T$, $\dot{\sigma}_4(0) = [0.1, 0.1, 0.1]^T$, respectively.

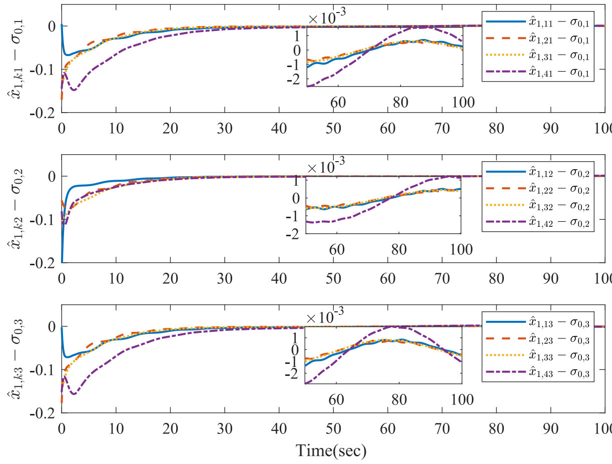

Fig. 3. Time histories of the attitude observation errors.

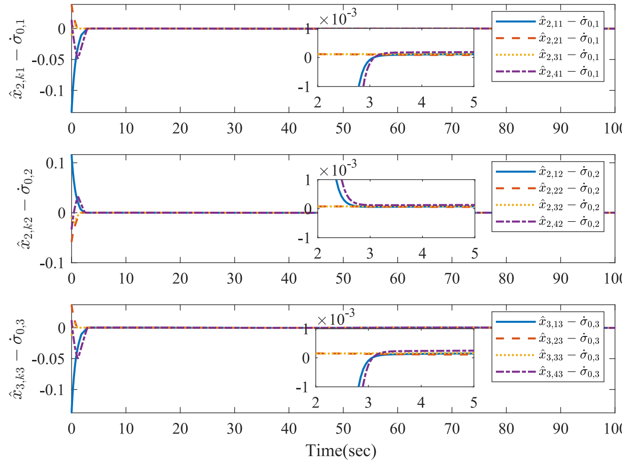

Fig. 4. Time histories of the angular velocity observation errors.

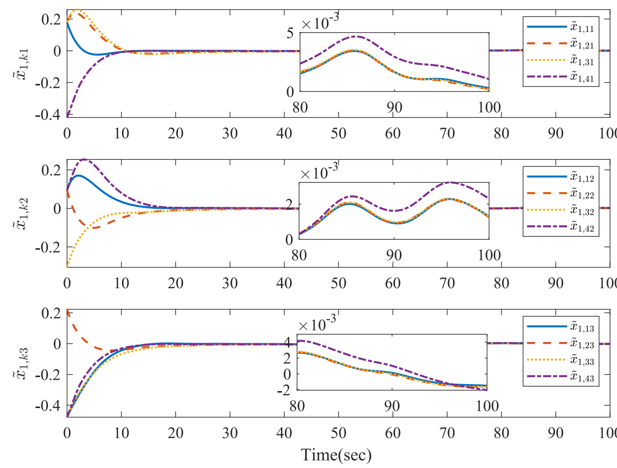

Fig. 5. Time histories of the attitude tracking errors.

The elements of actuator effectiveness matrix are selected as the ones in [18]. The external disturbance is chosen as $d_i = [2\cos(0.2\pi t) - \cos(0.4\pi t), 2\cos(0.2\pi t) - 2\cos(0.4\pi t), \sin(0.2\pi t) - 2\cos(0.4\pi t)\ ]^T \times 10^{-2}$Nm, $i = 1, 2, 3, 4$. The primary weights are selected as 1 if there exists an edge between two spacecrafts, and the unknown faults on the communication link are assumed to be $f_{a_{ij}}(t) = 0.5\sin(t) \times \text{rand} + 0.001t \times \text{rand} + 0.01r(t) \times \text{rand}$ and $f_{b_i}(t) = 0.5\cos(t) \times \text{rand} + 0.001t \times \text{rand} + 0.01r(t) \times \text{rand}$, where $\text{rand}(\cdot) \in (0, 1)$ being a random number generator and $r(t) = \{1,\ t > T\ or\ 0,\ t \le T\}$ with T being a designed step time. The state of leader spacecraft, which is also the desired command, is selected as $\sigma_0 = 0.15[\sin(0.09t), \sin(0.08t), \sin(0.1t)]^T$.

Based on the results of Theorems 1 and 2, the control parameters are chosen as follows($i = 1, \ldots, 5, k = 1, 2, 3$): $c_{1,i}(0) = 1$, $c_{2,i}(0) = 0.5$, $\Upsilon_{ik}(0) = 0.2$, $k = 1.5$, $l_1 = 0.025$, $l_2 = 0.05$, $\alpha = 0.9$, $\beta = 1.1$, $r_1 = 0.7$, $p_1 = 1.5$, $p_2 = 2$, $\delta = 1.2$, $\theta = 0.8$, $k_1 = 0.2$, $k_2 = 5$, $k_3 = 0.1$, $k_4 = 10$, $k_5 = 0.1$, and $k_6 = 10$. Besides, for the NN function, the number of nodes is selected as 10, the centers μ_i are selected as $-0.5 + 0.1i$, and the widths are selected as $\delta_i = 2$, for $i = 1, 2, \ldots, 10$.

For illustrating the performance of the developed resilient fixed-time control scheme [the observer (8) and the controller (38)], two sets of numerical simulations are conducted.

Case A: With the developed observer (8), the time histories of the attitude and angular velocity observation errors of the follower spacecraft are recorded in Figs. 3 and 4, respectively. As shown in these figures, we can conclude that: the observation errors of the follower spacecraft converge to some small neighborhoods around zero ($|\hat{x}_{1,ki} - \sigma_{0,1}| \le 3 \times 10^{-3}$ and $|\hat{x}_{2,ki} - \dot{\sigma}_{0,1}| \le 1 \times 10^{-3}$) with the developed observer within about 80s, which implies that the observation is well achieved and the proposed observer is resilience with communication link faults by applying the adaptive gain technique.

Case B: With the proposed observer (8) and controller (38), the time histories of the attitude and angular velocity tracking errors are recorded in Figs. 5 and 6, respectively. According to these figures, we can obtain that these errors of the follower spacecraft converge to some small neighborhoods around zero ($|\tilde{x}_{1,ki}| \le 5 \times 10^{-3}$ and $|\tilde{x}_{2,ki}| \le 5.5 \times 10^{-4}$) with the developed control scheme within about 95s, which means that the proposed method is resilient to communication link faults, actuator faults, inertial uncertainties, and external disturbances. The time histories of the control torque are depicted in Fig. 7. It shows that the input signals are not only smooth but also in a reasonable range ($<$ 6 Nm) throughout the whole time interval.

To further demonstrate the superiority of the developed scheme, the traditional finite-time control strategy presented in [10] is demonstrated for the purpose of the comparison. For this aim, the control performance indices $\text{CI}_1 = \sqrt{\sum_{i=1}^{4} \|\tilde{x}_{1,i}\|^2}$ and $\text{CI}_2 = \sqrt{\sum_{i=1}^{4} \|\tilde{x}_{2,i}\|^2}$ are considered. Given the same initial conditions, the responses of different control schemes are recorded in Fig. 8. It is found that the developed fixed-time method has a faster convergence rate and a better-steady-state performance than the finite-time one. Furthermore, the control effort of both schemes is recorded in Fig. 9. Compared with the finite-time control scheme, the developed scheme has larger energy consumption during the transient phase from 0-5 s and smaller energy consumption during the steady-state phase. Overall, the proposed control scheme remains a similar control effort as the finite-time one.

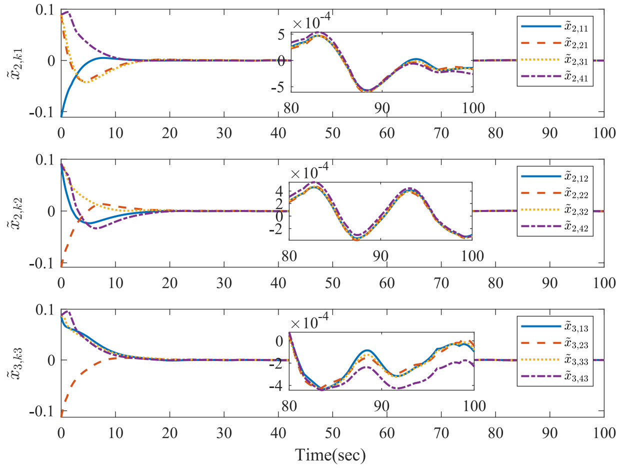

Fig. 6. Time histories of the angular velocity tracking errors.

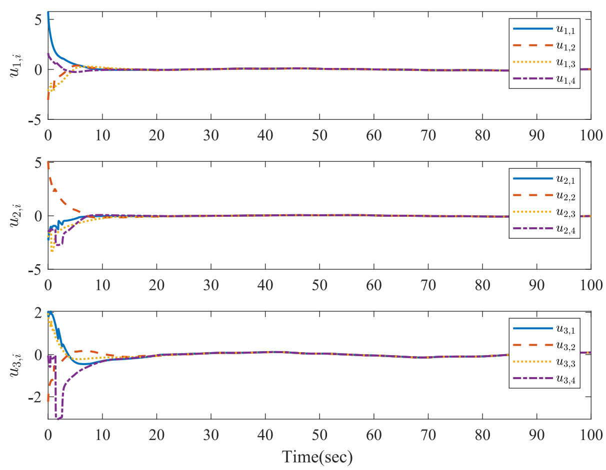

Fig. 7. Time histories of the control input signals.

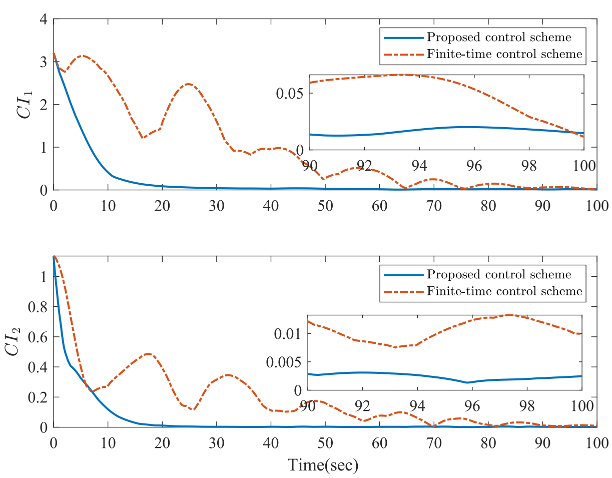

Fig. 8. Time histories of the control performances.

In other words, the better-control performance and greater resilience of the developed method do not depend on the extra control effort. Therefore, the proposed control approach is an effective one.

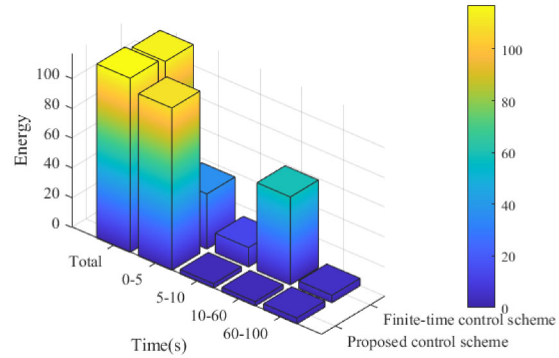

Fig. 9. Energy comparison between the proposed control scheme and finite-time one.

V. Conclusion

In this work, a resilient neuroadaptive distributed fixed-time attitude coordination control scheme has been presented by means of the adaptive method, NN approximation, and adding a power integrator technique. The developed method achieves the resilient attitude coordination control under the directed communication graph, despite the presence of communication link faults, actuator faults, inertial uncertainties, and external disturbances; moreover, it can guarantee that the attitude tracking errors converge to a small neighborhood of the origin in a fixed time and does not require the complex computation. The simulation results demonstrate the effectiveness of the developed method. Notably, the developed control scheme can also be utilized to solve the other nonlinear formation control problems. Our future research effort will be dedicated toward investigating the resilient event-triggered attitude coordination control for SFF.

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